ENEE12016 - Signals and Systems

Lecture Notes

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Contents

1 Signals and Times Systems **6**

1.1 Signals: . . . 6
1.2 Continuous vs. Discrete: . . . 6
1.3 Sampling: . . . 7
1.4 Nyquist Theorem: . . . 7
1.5 System: . . . 7
1.6 Signal Transformation: . . . 7
1.6.1 Amplitued Transform . . . 7
1.6.2 Amplitude Shift . . . 8
1.6.3 Time Reversal . . . 8
1.6.4 Time Scaling . . . 8
1.6.5 Time Shifting . . . 8
1.7 Even and Odd Signals: . . . 8
1.8 Periodic Signals: . . . 9
1.8.1 Complex Form . . . 9
1.8.2 Fundamental Period . . . 9
1.9 General Complex Exponential: . . . 10
1.10 Step Function: . . . 10
1.11 Impulse Function: . . . 10
1.12 Signal Energy and Power: . . . 10
1.12.1 Continuous . . . 10
1.12.2 Discrete . . . 11
1.12.3 Fundamentals . . . 11
1.13 System Representation: . . . 11
1.14 Causality: . . . 11
1.15 Invertability: . . . 11
1.15.1 Invertable . . . 12
1.15.2 Non-Invertable . . . 12

2 Linear Time Invariant Systems **12**

2.1 What are They?: . . . 12
2.2 Time Invarience: . . . 12
2.3 Discrete Signals as Impulse Functions: . . . 13
2.4 Convolution of Discrete Signals: . . . 13
2.5 Convolution Sum-Table Method: . . . 14
2.6 Convolution Sums Graphical Representation: . . . 15
2.7 Continuous Time Signals: . . . 15
2.8 Pulse Function: . . . 15
2.9 Continuous Time Series Approximation: . . . 15

2.10 Pulse Response: 16

3 Laplace Transformation **16**
3.1 Why Laplace?: 16
3.2 Laplace Transform: 16
3.3 Conventions: 17
3.4 Integration by Parts: 17
3.5 Operational Transforms: 19
3.5.1 Multiplication by a Constant 19
3.5.2 Addition and Subtraction 19
3.5.3 Differentiation 19
3.5.4 Integration 20
3.5.5 Translation in Time Domain 20
3.5.6 Translation in Frequency Domain 20
3.5.7 Scale Change 20
3.5.8 Frequency Differentiation 21
3.6 Inverse Transform: 21
3.7 Partial Fraction Expansion: 21

4 s-Domain Analysis **21**
4.1 Circuit Elements in s-Domain: 21
4.1.1 Resistor 22
4.1.2 Inductor 23
4.1.3 Capacitor 23
4.2 Circuit Analysis: 23

5 Transfer Functions **24**
5.1 Complex Frequency: 24
5.2 Pole Zero Diagram: 24
5.3 LTI Systems in Differentials: 25
5.4 Linear Constand Coefficient Differential Equations: 25
5.5 Transfer Function: 26
5.6 Poles and Zeros of Transfer Function: 26
5.7 Impulse Response: 27
5.8 Use of Convolution Integral: 27
5.9 Initial and Final Value Theorem: 28
5.9.1 Initial Value 28
5.9.2 Final Value 28
5.10 L'Hopital's Rule: 28
5.11 Complete System Response: 29

6 Stability of LTI Systems: 31
6.1 Region of Convergence of Laplace: . . . 31
6.2 Region of Convergence: . . . 32
6.3 System Stability: . . . 32
6.4 Bounded Input Bounded Output Stability: . . . 32
6.5 Discrete Signal Stability: . . . 34
6.6 System Stability - s-Domain: . . . 34
6.7 Location of Poles: . . . 35
6.8 Block Diagrams: . . . 35
6.9 Using Integrators: . . . 35
6.10 Feedback System: . . . 36
6.11 z-Transform: . . . 36

7 Fourier Series 37
7.1 Signals as Other Signals: . . . 37
7.2 Periodic Waveforms: . . . 37
7.3 Fourier Series: . . . 37
7.4 Dirichlet's Conditions: . . . 38
7.5 Signal Symmetry and Fourier Coefficients: . . . 38
7.5.1 Even . . . 38
7.5.2 Odd . . . 39
7.5.3 Non-symmetric . . . 39
7.6 Half-wave Symmetry: . . . 40
7.6.1 Even and Half-wave . . . 40
7.6.2 Odd and Half-wave . . . 41
7.7 Quater-wave Symmetry: . . . 41
7.7.1 Even and Quater-wave . . . 41
7.7.2 Odd and Quater-wave . . . 41
7.8 Steady-state Response: . . . 42
7.9 Fourier Series - Alternative Form: . . . 43
7.10 Average Power of Periodic Signal: . . . 43
7.11 RMS of Periodic Signals: . . . 44
7.12 Exponential Form of Fourier Series: . . . 44
7.13 Summary of Exponential Form: . . . 45

8 Fourier Transform 45
8.1 Fourier Series: . . . 45
8.2 Fourier Transform: . . . 46
8.3 Inverse Fourier Transform: . . . 47
8.4 Laplace vs. Fourier: . . . 47
8.5 Convergence of Fourier Integral: . . . 47

8.6 Fourier Transform of Periodic: 48
8.7 Operational Transforms: 48
8.7.1 Linearity 48
8.7.2 Differentiation 49
8.7.3 Integration 49
8.7.4 Translation in t-Domain 49
8.7.5 Translation in w-Domain 49
8.7.6 Scale Change 49
8.7.7 Convolution in t-Domain 49
8.7.8 Convolution in w-Domain 49
8.7.9 Modulation 49
8.8 Conjugate Symmetry: 50
8.9 Duality Principle: 51
8.10 Parseval's Theorem: 51

9 Frequency Selective Circuits **51**
9.1 Time vs. Frequency Domain: 51
9.2 Signal Transmission: 51
9.3 Circuit Components and Frequency: 51
9.4 Effects of Phase Changes: 52
9.5 Frequency Response of an LTI System: 52
9.6 Effects of Phase Response: 52
9.7 Bode Plot: 52
9.8 Frequency Selective Circuits: 52
9.9 Filter Categories - Ideal: 53
9.9.1 Low Pass 54
9.9.2 High Pass Filter 57
9.9.3 Band-pass Filter 58
9.9.4 Band-pass Characteristics 59
9.9.5 Band-reject Filter 61

10 Active Filter Design **63**
10.1 Filter Design: 63
10.2 Filter Gain: 63
10.3 Filter Order: 64
10.4 Active Filters: 64
10.4.1 First Order Active Low Pass 64
10.4.2 First Order Active High Pass 65
10.4.3 Active Bandpass Filter 66
10.4.4 Active Bandreject Filters 67
10.5 Cascading Filters: 68

10.6 n Order Low Pass Filter: 68
10.7 Unity Design and Scaling: 69
10.8 Filter Gain around Cutoff: 70
10.9 Butterworth Filter: 70

1 Signals and Times Systems

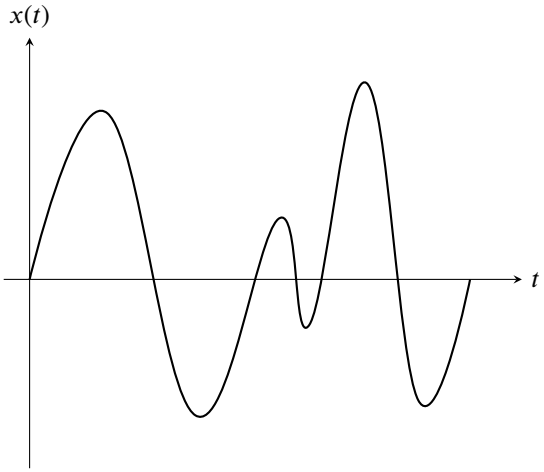
1.1 Signals:

A signal is a physical quantity that varies by something. The most common ones are time, space, and angle. Basically if it ca be measured or its effects measured, it can be though of as a signal. For example:

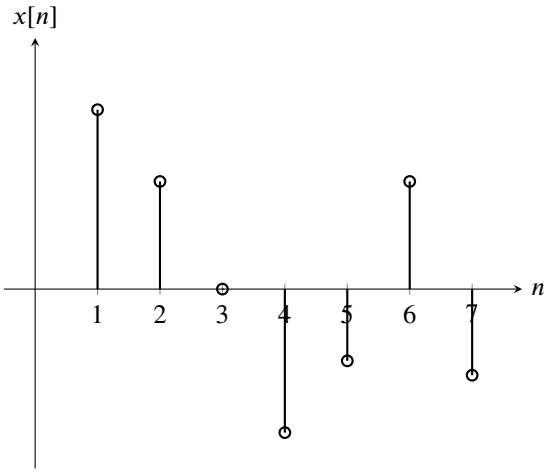
- Radio waves
- AC waveform
- Temperature
- Tide height
- Digital bit streams

1.2 Continuous vs. Discrete:

A continuous signal has a value for all time points. A waveform with considered infinite precision.



Discrete signalling on the other hand, only has values at set time points.



1.3 Sampling:

A discrete signal can be made by sampling a continuous signal of set time intervals. The sampled frequency must be twice the continuous frequency, as a minimum. The high the sampled frequency, the more accurate the output waveform.

1.4 Nyquist Theorem:

If a function $x(t)$ contains a highest frequency of BHz. For it too be completely determined sampling must be done at $\frac{1}{2B}$s.

1.5 System:

A group of regular items interacting to form a while. Such as, power amp, transformer, or laptop.

1.6 Signal Transformation:

1.6.1 Amplitued Transform

Scaling the signal

$$x(t) \rightarrow ax(t)$$

1.6.2 Amplitude Shift

Vertical shift

$$x(t) \rightarrow x(t) + b$$

1.6.3 Time Reversal

A flip across the y-axis

$$x(t) \rightarrow x(-t)$$

1.6.4 Time Scaling

Changing the frequency

$$x(t) \rightarrow x(bt)$$

- If $b > 1$: Compressing the signal
- Else $0 < b < 1$: Expanded the signal

1.6.5 Time Shifting

Horizontal movement

$$x(t) \rightarrow x(t - t_0)$$

- If $t_0 > 0$: Shifts signal right
- Else $t_0 < 0$: Shifts signal left

1.7 Even and Odd Signals:

Even signals are symmetric around the origin.

$$x(t) = x(-t)$$

Odd signals are asymmetric.

$$x(t) = -x(-t)$$

Any signal can be decomponsed into even and odd components.

$$x_e(t) = \frac{1}{2}[x(t) + x(-t)]$$

$$x_o(t) = \frac{1}{2}[x(t) - x(-t)]$$

Sub $t = -t$, into even components.

$$\begin{aligned} x_e(-t) &= \frac{1}{2}[x(-t) + x(-(-t))] \\ &= \frac{1}{2}[x(t) + x(-t)] \\ &= x_e(t) \\ &\therefore \text{Even} \end{aligned}$$

Sub $t = -t$, into odd components.

$$\begin{aligned} x_o(-t) &= \frac{1}{2}[x(-t) - x(-(-t))] \\ &= \frac{1}{2}[-x(t) + x(-t)] \\ &= -\frac{1}{2}[x(t) - x(-t)] \\ &= -x_o(t) \\ &\therefore \text{Odd} \end{aligned}$$

$$x(t) = x_e(t) + x_o(t)$$

1.8 Periodic Signals:

$$x(t) = x(t + T_0),\ T_0 > 0$$

This is where T_0 is the period.

1.8.1 Complex Form

$$\begin{aligned} x(t) &= e^{j\omega_0(t+T_0)} \\ &= e^{j\omega_0 t} e^{j\omega_0 T_0} \end{aligned}$$

1.8.2 Fundamental Period

$$T_0 = \frac{2\pi}{\omega_0}$$

1.9 General Complex Exponential:

$$C = |C|e^{j\phi}, \quad a = r + j\omega$$

$$\begin{aligned} x(t) &= |C|e^{j\phi(r+j\omega_0)t} \\ &= |C|e^{rt}e^{j\phi}e^{j\omega_0 t} \qquad = |C|e^{rt}\cos(\omega_0 t + \phi) + j|C|e^{rt}\sin(\omega_0 t + \phi) \end{aligned}$$

1.10 Step Function:

Switching operations require abrupt changes in $v(t)$ and t. At the switching point, it is undefined.

$$Ku(t) = 0,\ t < 0$$
$$Ku(t) = K,\ t > 0$$
$$Ku(t) = und\,f,\ t = 0$$

1.11 Impulse Function:

This function has infinite magnitude and zero duration. No practical application contains this, as nothing last with zero time. It is used to define derivative at discontinuity point. Discrete signals can be saved to be made up of multiple impulse functions.

$$x[n] = \sum_{k=-\inf}^{k=\inf} x[k]\delta[n-k]$$

1.12 Signal Energy and Power:

1.12.1 Continuous

$$E = \int_{t_1}^{t_2} |x(t)|^2 dt$$

$$P = \frac{1}{t_2 - t_1}\int_{t_1}^{t_2} |x(t)|^2 dt$$

$$E_\infty = \lim_{T\to\infty}\int_{-T}^{T} |x(t)|^2 dt$$

$$P_\infty = \lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T} |x(t)|^2 dt$$

1.12.2 Discrete

$$E = \sum_{n_1}^{n_2} |x[n]|^2$$

$$P = \frac{1}{n_2 - n_1} \sum_{n_1}^{n_2} |x[n]|^2$$

$$E_\infty = \lim_{N \to \infty} \sum_{-N}^{N} |x[n]|^2$$

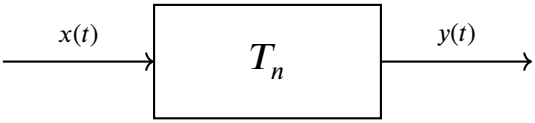
$$P_\infty = \lim_{N \to \infty} \frac{1}{2N+1} \sum_{-N}^{N} |x[n]|^2$$

1.12.3 Fundamentals

If energy over an infinite interval is finite, $E_\infty < \infty$, then P_{av} is 0, $P_\infty = 0$. Else $P_\infty > 0 \quad \rightarrow \quad E_\infty = \infty$. A few cases can have $E_\infty = \infty$ and $P_\infty = \infty$. One such one is $x(t) = t$.

1.13 System Representation:

A system is usually represented by a black box.



$$y(t) = T_n[x(t)]$$

The T_n is not a function but a transformation happening to the function.

1.14 Causality:

A system is casual if it only depends on present and past values.

$$y(t) = 5x(t) = 6x^2(t-1) + y(t-1)$$

1.15 Invertability:

A system is said to be invertible if the operations placed on it can be revered, giving a unique answer.

1.15.1 Invertable

$$y[n] = 2x[n]$$
$$x[n] = \frac{1}{2}y[n]$$

$$y(t) = \int_{-\infty}^{t} x(\tau)d\tau$$
$$x(t) = \frac{dy(t)}{dt}$$

1.15.2 Non-Invertable

$$y(t) = x^2(t)$$
$$x(t) = \pm\sqrt{y(t)}$$

Produces 2 possible results

$$y[n] = nx[n]$$
$$x[n] = \frac{1}{n}y[n]$$

Non result if $n \neq 0$

2 Linear Time Invariant Systems

2.1 What are They?:

A linear system is one which obeys the laws of linearity and superposition:

- If x is multiplied by k making kx, $f(x)$ is also now $kf(x)$.
- When more than one factor is applied it is the sum of individual responses to each input.

2.2 Time Invarience:

Behaviour and characteristic of the system is fixed over time. An example of this is an RC circuit characteristics will not change if the experiment is conducted today, tomorrow or yesterday. Mathematically, if the input is shifted by

T, the output is also shifted by T.

$$x(t) \Rightarrow y(t)$$

$$x(t-T) \Rightarrow y(t-T)$$

2.3 Discrete Signals as Impulse Functions:

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$$

For any given value of n, only 1 impulse in non-zero, and the rest are zero. Assume the response of an LTI system to unit impulse $\delta[n]$ is $h[n]$. Thus $\delta[n-k]$ is $h[n-k]$.

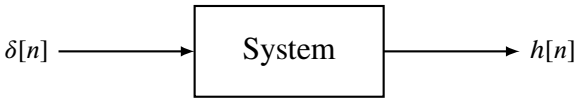
2.4 Convolution of Discrete Signals:

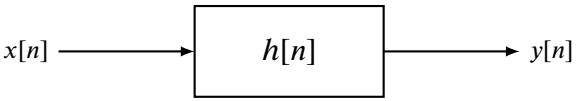
The response of that system to signal $n[n]$ is superposition of a scaled and time shifted $h[n]$.

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

This is often called the convoluetion sum.

$$\begin{aligned} y[n] &= x[n] \otimes h[n] \\ &= \sum_{k=-\infty}^{\infty} x[k]h[n-k] \end{aligned}$$





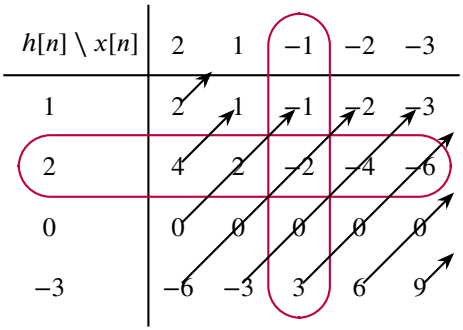
2.5 Convolution Sum-Table Method:

Let:

$$x[n] = 2, 1, -1, -2, -3$$

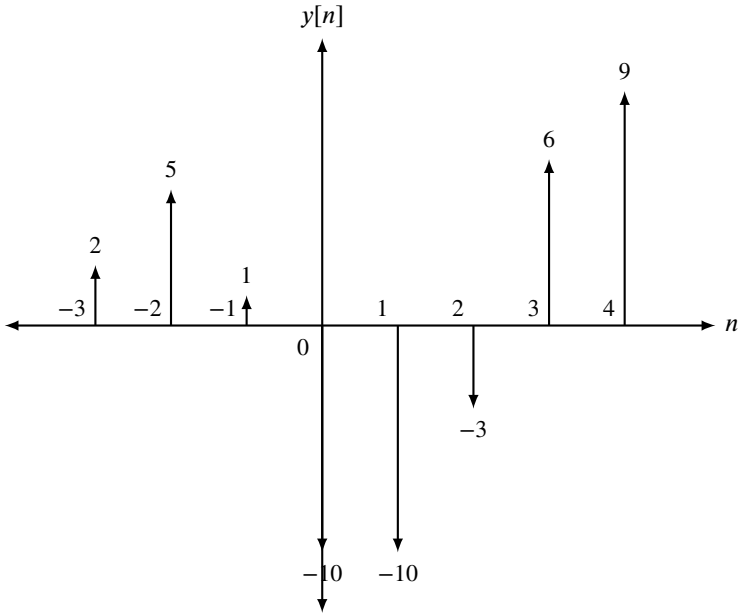
$$h[n] = 1, 2, 0, -3$$

$$x[0] = -1, \text{ and } h[0] = 2$$



$$y[n] = 2, 4, 1, -10, -10, -3, 6, 9$$

$$y[0] = -10$$



2.6 Convolution Sums Graphical Representation:

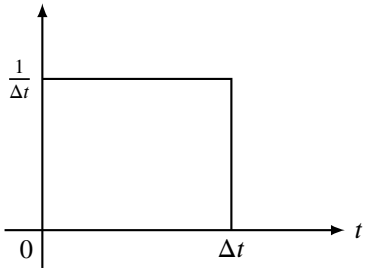
- Draw input signal as a function of k
- Draw impulse responses as function of k
- Time reverse the impulse response
- Shift impulse along t-axis
- For each n take the sum of product between input and shifted overlaps

2.7 Continuous Time Signals:

Any continuous time signal can be approximated by a thin set of strips, known as staircase approximations. The height of the strip, $\hat{x}(t)$, is the value of $x(t)$ at the start of the strip.

2.8 Pulse Function:

$$p(t) = \left\{ \begin{array}{l} \frac{1}{\Delta t},\ 0 < t < \Delta t \\ 0,\ \text{Elsewhere} \end{array} \right\}$$



$$\int_{-\infty}^{\infty} p(t)dt = t$$

2.9 Continuous Time Series Approximation:

This can be approximated as a series of pulse functions.

- Pulse located at integral multiples of pulse width (Δt)
- Pulse height is $x(\Delta t)$

Pulse of $t = \Delta t$.

$$x(k\Delta t)[p(t - k\Delta t) \cdot \Delta t] = x(\Delta t)$$

The signal can then be approximated as the sum of these signals.

$$\hat{x}(t) = \sum_{k=-\infty}^{\infty} x(k\Delta t)[p(t - k\Delta t) \cdot \Delta t]$$

As $\Delta t \to 0$, $p(t)$ approaches an impulse function.

$$x(t) = \int_{\infty}^{\infty} x(\tau)\delta(t - \tau)d\tau$$

As $\Delta t \to 0$, $k\Delta t \to \tau$, $\Delta t \to d\tau$ and $p(t - k\Delta t) \to \delta(t - \tau)$.

2.10 Pulse Response:

If $h(t)$ is the response to the LTI system, $h_p(t)$ can be defined to be the response of a pulse system.

3 Laplace Transformation

3.1 Why Laplace?:

- Solving linear circuits frequently involves solving linear differential equations. These include transient analysis, AC analysis and transient or steady state system response.
- Finding time domain convolution
- Signals with many freqency components may be more apparent in a different domain

3.2 Laplace Transform:

$$\mathcal{L}\{f(t)\} = F(s) = \int_{0}^{\infty} f(t)e^{-st}dt$$

- Convert a time domain function to frequency domain. $t \Rightarrow s,$ where $s = \sigma + j\omega$
- Important Notes
  - If intergrand does not converge there is no Laplace transform
  - Only currently $t > 0^{-}$

3.3 Conventions:

Time domain		Frequency Domain
$f(t)$	$\Rightarrow$	$F(s)$
$g(t)$	$\Rightarrow$	$G(s)$

Functional transform is only specific functions. Operation transofmr is the description of how mathematical operations are done using Laplace.

3.4 Integration by Parts:

$$\int_0^\infty uv'dt = [uv]_0^\infty - \int_0^\infty u'vdt$$

Make sure that u simplifies when derived and v' simplifies when integrated. Laplace transform of 1:

$$\begin{aligned}
\mathcal{L}\{f(t)\} = F(s) &= \int_0^\infty f(t)e^{-st}dt \\
\mathcal{L}\{1\} &= \int_0^\infty e^{-st}dt \\
&= [-\frac{1}{s}e^{-st}]_0^\infty \\
&= [0] - [-\frac{1}{s}] \\
&= \frac{1}{s}
\end{aligned}$$

$$\begin{aligned}
f(t) &\Leftrightarrow F(s) \\
1 &\Leftrightarrow \frac{1}{s}
\end{aligned}$$

Laplace transform of t:

$$\mathcal{L}\{t\} = F(s) = \int_0^\infty te^{-st}dt$$

Let $u = t, \quad v' = e^{-st}$

$$u' = 1, \quad v = [-\frac{1}{s}e^{-st}]_0^\infty$$

$$\begin{aligned} F(s) &= uv - \int u'vdt \\ &= [t(-\frac{1}{s})]_0^\infty - \int_0^\infty -\frac{1}{s}e^{-st}dt \\ &= 0 + \frac{1}{s}\int_0^\infty e^{-st}dt \\ &= \frac{1}{s}[-\frac{1}{s}e^{-st}]_0^\infty \\ &= -\frac{1}{s^2}[0-1] \\ &= \frac{1}{s^2} \end{aligned}$$

$$t^n \Leftrightarrow \frac{n!}{s^{n+1}}$$

Laplace transform of $\delta(t)$:

$$\begin{aligned} \mathcal{L}\{(\delta(t)\} = F(s) &= \int_{0^-}^\infty \delta(t-0^-)e^{-st}dt \\ &= e^{-s\cdot 0} \\ &= 1 \end{aligned}$$

$$\delta(t) \Leftrightarrow 1$$

In general

$$\mathcal{L}\{\frac{d^n[\delta(t)]}{dt^n}\} = s^n$$

3.5 Operational Transforms:

3.5.1 Multiplication by a Constant

$$\begin{aligned}\mathcal{L}\{f(t)\} &= F(s) = \int_0^\infty f(t)e^{-st}dt \\ \mathcal{L}\{Kf(t)\} &= \int_0^\infty Kf(t)e^{-st}dt \\ &= K\int_0^\infty f(t)e^{-st}dt \\ &= KF(s)\end{aligned}$$

3.5.2 Addition and Subtraction

$$\begin{aligned}\mathcal{L}_1\{f_1(t)\} &= F_1(s) \\ \mathcal{L}_2\{f_2(t)\} &= F_2(s)\end{aligned}$$

$$\begin{aligned}\mathcal{L}\{f_1(t) \pm f_2(t)\} &= \int_0^\infty (f_1(t) \pm f_2(t))e^{-st}dt \\ &= \int_0^\infty f_1(t)e^{-st}dt \pm \int_0^\infty f_2(t)e^{-st}dt \\ &= F_1(s) \pm F_2(s)\end{aligned}$$

3.5.3 Differentiation

$$\mathcal{L}\{\frac{df(t)}{dt}\} = \int_{0^-}^\infty \frac{df(t)}{dt}e^{-st}dt$$

Let: $u' = \frac{df(t)}{dt}$ $\quad v = e^{-st}$

$u = f(t)$ $\quad v' = -se^{-st}$

$$\begin{aligned}\mathcal{L}\{\frac{df(t)}{dt}\} &= uv - \int uv' \\ &= [f(t)e^{-st}]_{0^-}^\infty - \int -sf(t)e^{-st}dt \\ &= \int_{0^-}^\infty f(t)se^{-st}dt - f(0^-) \\ &= sF(s) - f(0^-)\end{aligned}$$

In General:

$$\mathcal{L}\{\frac{d^n f(t)}{dt^n}\} = s^n F(s) - s^{n-1} f(0^-) - s^{n-2}\frac{df(0^-}{dt} - s^{n-3}\frac{d^2 f(0^-)}{dt^2}$$
$$- \ldots - \frac{d^{n-1} f(0^-)}{dt^{n-1}}$$

3.5.4 Integration

$$\mathcal{L}\{\int_{0^-}^{t} f(x)dx\} = \int_{0^-}^{\infty}[\int_{0^-}^{t} f(x)dx]e^{-st}dt$$
$$\text{Let: } u' = e^{-st} \quad v = \int_{0^-}^{t} f(x)dx$$
$$u = -\frac{1}{s}e^{-st} \quad v' = f(t)$$
$$\mathcal{L}\{\int_{0^-}^{t} f(x)dx\} = [-\frac{e^{-st}}{s}\int_{0^-}^{t} f(x)dx]_{0^-}^{\infty} - \int_{0^-}^{\infty} -f(t)\frac{e^{-st}}{s}dt$$
$$\lim_{t\to\infty} -\frac{e^{-st}}{s} \to 0$$
$$\lim_{t\to 0^-}\int_{0^-}^{t} f(x)dt \to 0$$
$$\therefore \mathcal{L}\{\int_{0^-}^{t}\} = \int_{0^-}^{\infty} f(t)\frac{e^{-st}}{s}dt$$
$$= \frac{1}{s}\int_{0^-}^{\infty} f(t)e^{-st}dt$$
$$= \frac{F(s)}{s}$$

3.5.5 Translation in Time Domain

$$\mathcal{L}\{f(t-a)u(t-a)\} = e^{-as}F(s)$$

3.5.6 Translation in Frequency Domain

$$\mathcal{L}\{e^{-at}f(t)\} = F(s+a)$$

3.5.7 Scale Change

$$\mathcal{L}\{f(at)\} = \frac{1}{a}F(\frac{s}{a})$$

3.5.8 Frequency Differentiation

$$\mathcal{L}\{tf(t)\} = -\frac{dF(s)}{ds}$$

$$\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n F(s)}{ds^n}$$

3.6 Inverse Transform:

Once the time domain problem is solved using Laplace transformation, it needs to be converted back to time domain to understand what has happenend.

$$F(s) = \frac{N(s)}{D(s)} = \frac{a_n s^n + a_{n-1}s^{n-1} + \ldots + a_1 s^1 + a_0 s^0}{b_m s^m + b_{m-1}s^{m-1} + \ldots + b_1 s^1 + b_0 s^0}$$

As such, $F(s)$ is needed to be converted back into a format which contains Laplace transformations of known functions.

3.7 Partial Fraction Expansion:

If $m > n$, a proper rational function expansion. Else $m \leq n$, is improper rational functions.

$$\begin{aligned} F(s) = \frac{N(s)}{D(s)} &= \frac{s+6}{s^4 + 5s^3 + 7s^2 + 3s} \\ &= \frac{s+6}{s(s+3)(s+1)^2} \\ &= \frac{K_1}{s} + \frac{K_2}{s+3} + \frac{K_3}{s+1} + \frac{K_4}{(s+1)^2} \end{aligned}$$

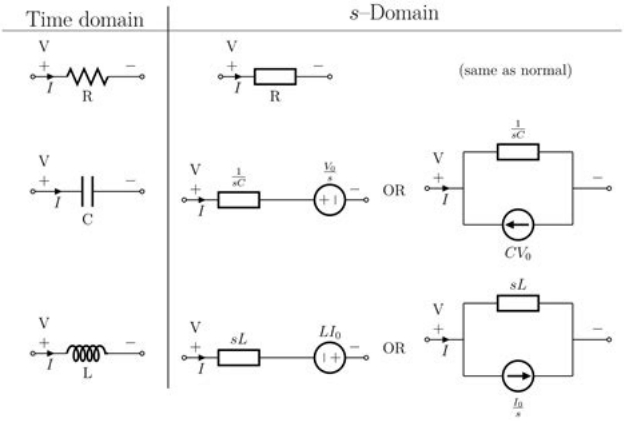
$$Ke^{-at}u(t) \Leftrightarrow \frac{K}{s+a}$$

$$f(t) = K_1 u(t) + K_2 e^{-3t}u(t) + K_3 e^{-t}u(t) + K_4 te^{-t}u(t)$$

4 s-Domain Analysis

4.1 Circuit Elements in s-Domain:

Similar to analysis done on AC circuits in time domain. Using Laplace transforms, the time domain formulas can be converted to s-Domain. They key difference is that the initial conditions are taken into during the conversion.



4.1.1 Resistor

$$
\begin{aligned}
v(t) &= Ri(t) \\
\mathcal{L}\{v(t)\} = V(s) &= \int_0^\infty v(t)e^{-st}dt \\
&= \int_0^\infty Ri(t)e^{-st}dt \\
&= R\int_0^\infty i(t)e^{-st}dt \\
\therefore V(s) &= RI(s) \\
I(s) &= \frac{V(s)}{R}
\end{aligned}
$$

The units change in s-Domain to volt-seconds and amp-seconds.

4.1.2 Inductor

$$
\begin{aligned}
v(t) &= L\frac{di(t)}{dt} \\
\mathcal{L}\{v(t)\} &= V(s) = \int_0^\infty L\frac{di(t)}{dt}e^{-st}dt \\
&= L\int_0^\infty \frac{di(t)}{dt}e^{-st}dt \\
&= L(sI(s) - i(0^-)) \\
\therefore V(s) &= sLI(s) - LI_0 \\
I(s) &= \frac{V(s)}{sL} + \frac{I_0}{s}
\end{aligned}
$$

The LI_0 is the initial conditions of the inductor at $t = 0^-$. LI_0 is a voltage sourse in series when finding voltage in s-Domain. $\frac{I_0}{s}$ is a current source in parrallel.

4.1.3 Capacitor

$$
\begin{aligned}
i(t) &= C\frac{dv(t)}{dt} \\
\mathcal{L}\{i(t)\} &= I(s) = \int_0^\infty C\frac{dv(t)}{dt}e^{-st}dt \\
&= C\int_0^\infty \frac{dv(t)}{dt}e^{-st}dt \\
&= C(sV(s) - v(0^-)) \\
\therefore I(s) &= sCV(s) - CV_0 \\
V(s) &= \frac{I(s)}{sC} + \frac{V_0}{s}
\end{aligned}
$$

V_0 is the voltage at $t = 0^-$.

4.2 Circuit Analysis:

All of Kirchoff's Laws (KCL and KVL) that applied in time domain, apply in s-Domain. The only important part is to have everything in one domain before analysis. The R, sL, and $\frac{1}{sC}$ are the s-Domain impedances for resistors, inductors and capacitors respectively.

5 Transfer Functions

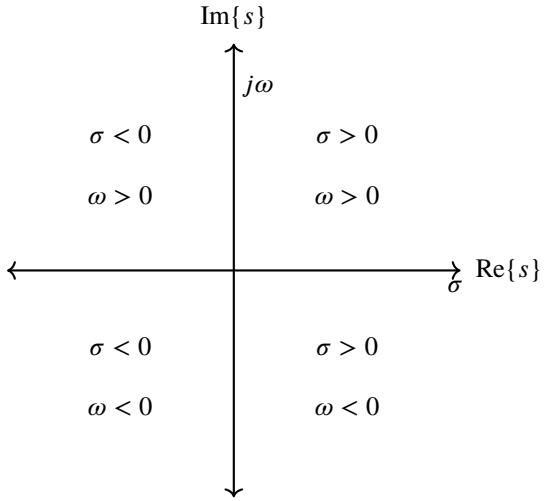
5.1 Complex Frequency:

In Laplace, it is taken in s-domain analysis, or complex frequency.

$$s = \sigma + j\omega$$

$$e^s = e^{\sigma + j\omega} = e^{\sigma} e^{j\omega}$$

This expansion produces a sinusoidal damped function. Using this, the s-plane can be identified.



5.2 Pole Zero Diagram:

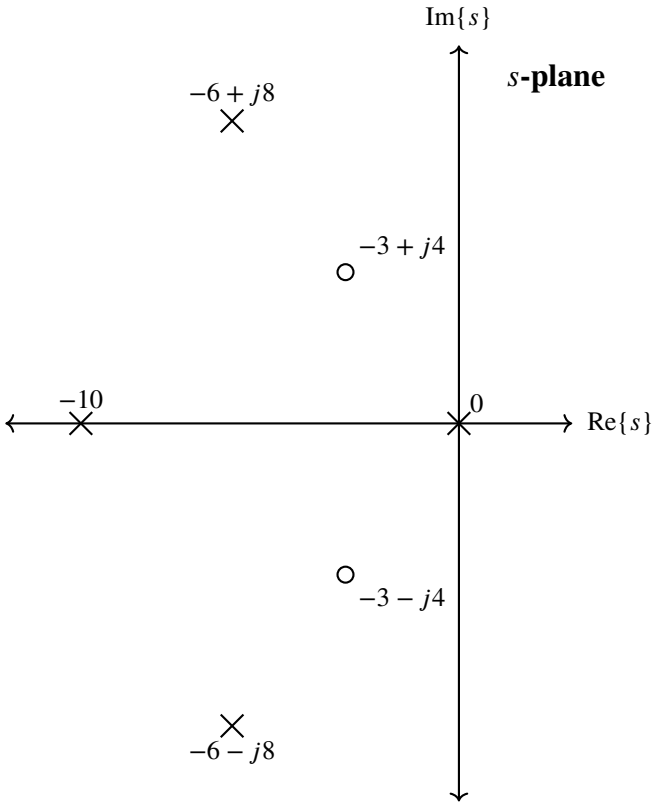
When analysis on 3D functions they are calculation intensive and often hard to visualize. Instead shown the extreme ends of functions mangnitude:

- Where Function is 0 - Zeros (Numerator is 0)
- Where Function is ∞ - Poles (Denominator is 0)

$$F(s) = \frac{10(s+5)(s+3-j4)(s+3+j4)}{s(s+10)(s+6-j8)(s+6+j8)}$$

$$0 \text{ at: } -5,\ -3+j4,\ -3-j4$$

$$\infty \text{ at: } 0,\ 10,\ -6+j8,\ -6-j8$$



5.3 LTI Systems in Differentials:

For example RLC circuits are discribed with differential equations, such as charging a capacitor:

$$C\frac{dv}{dt} + \frac{V}{R} = I_s$$

Parallel RLC Circuit:

$$\frac{d^2v}{dt^2} + \frac{1}{RC}\frac{dv}{dt} + \frac{1}{LC}v = 0$$

5.4 Linear Constand Coefficient Differential Equations:

$$\sum_{k=0}^{N} a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^{M} b_k \frac{d^k x(t)}{dt^k}$$

- $k = 0$, results in $x(t)$ and $y(t)$
- Order of the system is given by the highest derivative of $y(t)$
- This equation does not give an explicit expression of system output
- For $N = 0$, an explicit relationship of $y(t)$

$$y(t) = \frac{1}{a_0} \sum_{k=0}^{M} b_k \frac{d^k x(t)}{dt^k}$$

5.5 Transfer Function:

Taking Laplace of generic formula.

$$\mathcal{L}\{\frac{d^k y(t)}{dt^k}\} = s^k Y(s) - s^{k-1} y(0) - s^{k-2} \frac{dy(0)}{dt} - s^{k-3} \frac{d^2 y(0)}{dt^2} - \ldots - \frac{d^{k-1} y(0)}{dt^{-1}}$$

If $y(0) = 0$, then:

$$\mathcal{L}\{\frac{d^k y(t)}{dt^k}\} = s^k Y(s)$$

Take Laplace of both sides:

$$\mathcal{L}\{\sum_{k=0}^{N} a_k \frac{d^k y(t)}{dt^k}\} = \mathcal{L}\{\sum_{k=0}^{M} b_k \frac{d^k y(t)}{dt^k}\}$$

Assuming initial conditions are 0:

$$\sum_{k=0}^{N} a_k s^k Y(s) = \sum_{k=0}^{M} b_k s^k Y(s)$$

$$H(s) = \frac{Y(s)}{X(s)} = \frac{b_m s^m + b_{m-1} s^{m-1} + \ldots + b_1 s + b_0}{a_n s^n + a_{n-1} s^{n-1} + \ldots + a_1 s + a_0}$$

In general for any system the transfer function is the ratio of output to input.

5.6 Poles and Zeros of Transfer Function:

Any rational function, $H(s)$, can be expressed as the ratio of 2 polynomials.

$$H(s) = \frac{K(s + z_1)(s + z_2) + \ldots + (s + z_m)}{(s + p_1)(s + p_2) + \ldots + (s + p_n)}$$

From thos form, the poles and zeros can be found.

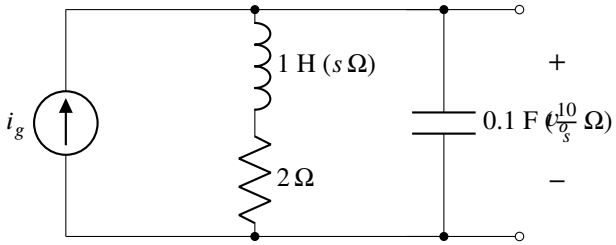
5.7 Impulse Response:

Recall that $\mathcal{L}\{\delta(t)\} = \Delta(s) = 1$. As such $Y(s) = H(s)$.

5.8 Use of Convolution Integral:

$$\begin{aligned} y(t) &= x(t) \otimes h(t) \\ &= \int_{\infty}^{\infty} h(t)x(t-\tau)d\tau \\ &= \int_{\infty}^{\infty} x(t)h(t-\tau)d\tau \end{aligned}$$

If $x(t)$ and $h(t)$ are known and Laplace transformable, the transfer function can be utilized.



Find transfer function $\frac{V_o}{i_g}$, Assume no stored energy in circuit.

$$\begin{aligned} Z_{eq} &= \frac{1}{\frac{1}{s+2} + \frac{1}{\frac{10}{s}}} \\ &= \frac{10(s+2)}{s^2 2s + 10} \end{aligned}$$

$$\begin{aligned} V_0(s) &= i_g(s) \cdot Z_{eq}(s) \\ H(s) &= \frac{V_0(s)}{i_g(s)} = Z_{eq}(s) \end{aligned}$$

Zeros at: -2.
Poles at: $-1 + j3,\ -1 - j3$

5.9 Initial and Final Value Theorem:

Laplace can be use to find values of $f(t)$ at 0 and ∞. This is useful as there values can be had to determine dineetly.

5.9.1 Initial Value

$$\lim_{t\to 0^+} f(t) = \lim_{s\to\infty} sF(s)$$

Where:

- $f(t)$ does not contain an impulse function
- $f(0^+)$ exists
- $F(s)$ can be found

5.9.2 Final Value

$$\lim_{t\to\infty} f(t) = \lim_{t\to 0} sF(s)$$

Final value does not exist if:

- $sF(s)$ has poles on right side of plane - growing exponentially
- $sF(s)$ has conjugate pole on $j\omega$ axis - sinusodial function
- $sF(s)$ has poles on origin - t^n function

5.10 L'Hopital's Rule:

If $f(t)$ is not defined when $t \to a$ then:

$$f(a) = \lim_{t\to a} f(t) = \frac{n'(a)}{d'(a)}$$

If the first derivative of function cannot be evaluated, then the second derivative is taken. Repeat until resolved. Find initial and final values:

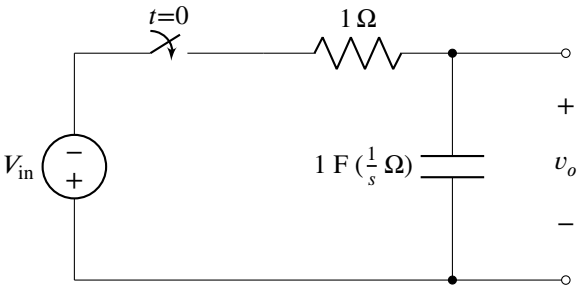
$$
\begin{aligned}
F(s) &= \frac{s^2 - 5s + 6}{s^3 + 4s^2 + 3s} \\
\lim_{t\to\infty} f(t) &= \lim_{s\to 0} sF(s) \\
&= \lim_{s\to 0} \frac{s(s^2 - 5s + 6)}{s^3 + 4s^2 + 3s} \\
&= \lim_{s\to 0} \frac{s^2 - 5s + 6}{s^2 + 4s^2 + 3} \\
&= \frac{6}{3} \\
&= 2
\end{aligned}
$$

$$
\begin{aligned}
\lim_{t\to 0^+} f(t) &= \lim_{s\to\infty} sF(s) \\
&= \lim_{s\to\infty} \frac{s^2 - 5s + 6}{s^2 + 4s + 3} \\
&\text{Using L'Hopital} \\
&= \lim_{s\to\infty} \frac{2s - 5}{2s + 4} \\
&\text{Using L'Hopital} \\
&= \lim_{s\to\infty} \frac{2}{2} \\
&= 1
\end{aligned}
$$

5.11 Complete System Response:

$$
\begin{aligned}
H(s) &= \frac{Y(s)}{X(s)} \\
\therefore Y(s) &= H(s)X(s)
\end{aligned}
$$

- Partial fraction expansion of $Y(s)$ includes terms from partial fractions (poles) of $H(s)$ and $Y(s)$
- Poles of $H(s)$ result in transient response of the system
- Poles of $X(s)$ results in steady-state response



$$\frac{V_0(s)}{V_{in}(s)} = H(s)$$

$$V_0(s) = \frac{\frac{1}{sC} \times V_{in}}{R + \frac{1}{sC}}$$

$$\frac{V_0(s)}{V_{in}(s)} = \frac{\frac{1}{s}}{1 + \frac{1}{s}}$$

$$= \frac{1}{1+s}$$

Response for V_{in}

$$V_0(s) = \frac{1}{s+1} V_{in}(s)$$

$$= \frac{1}{s+1} \cdot \frac{1}{s}$$

$$= \frac{1}{s(s+1)}$$

$$= \frac{K_1}{s} + \frac{K_2}{s+1}$$

$$K_1 = [\frac{1}{s+1}]_{s=0} = 1$$
$$K_2 = [\frac{1}{s}]_{s=1} = -1$$

$$\therefore V_0(s) = \frac{1}{s} + \frac{-1}{s+1}$$
$$\therefore v_0(t) = \mathcal{L}^{-1}\{V_0(s)\}$$
$$= (1 - e^{-t})u(t)$$

Response of $V_{in}(t) = \sin(t)$

$$\mathcal{L}\{V_{in}(t)\} = \frac{1}{s^2+1}$$
$$V_0(s) = \frac{1}{s+1} \times \frac{1}{s^2+1}$$
$$= \frac{1}{(s+1)(s-j)(s+j)}$$
$$= \frac{0.5}{s+1} + \frac{-0.25\sqrt{2}e^{j45^\circ}}{s-j} + \frac{-0.25\sqrt{2}e^{-j45^\circ}}{s+j}$$
$$\therefore v_0(t) = \mathcal{L}^{-1}\{V_0(s)\}$$
$$= (0.5e^{-t} - 0.5\sqrt{2}\cos(t - 45^\circ))u(t)$$

Transient Response: $0.5e^{-t}$
Steady-state Response: $0.5\sqrt{2}\cos(t - 45^\circ)$

6 Stability of LTI Systems:

6.1 Region of Convergence of Laplace:

- Some integrals only converge (exist) for a certain range
- The region where the Laplace transform converges is RoC
- This is expressed in a range of s-values and should be stated with the Laplace. Outside this range Laplace $\rightarrow \infty$

$$
\begin{aligned}
x(t) &= e^{-at}u(t) \\
X(s) &= \mathcal{L}\{x(t)\} \\
&= \int_0^\infty e^{-at}e^{-st}dt \\
&= \frac{1}{s+a} \\
&= \frac{1}{\sigma + j\omega + a} \\
s + a &= 0 \\
\therefore s &= -a
\end{aligned}
$$

RoC only considers $Re\{s\} = \sigma$.

6.2 Region of Convergence:

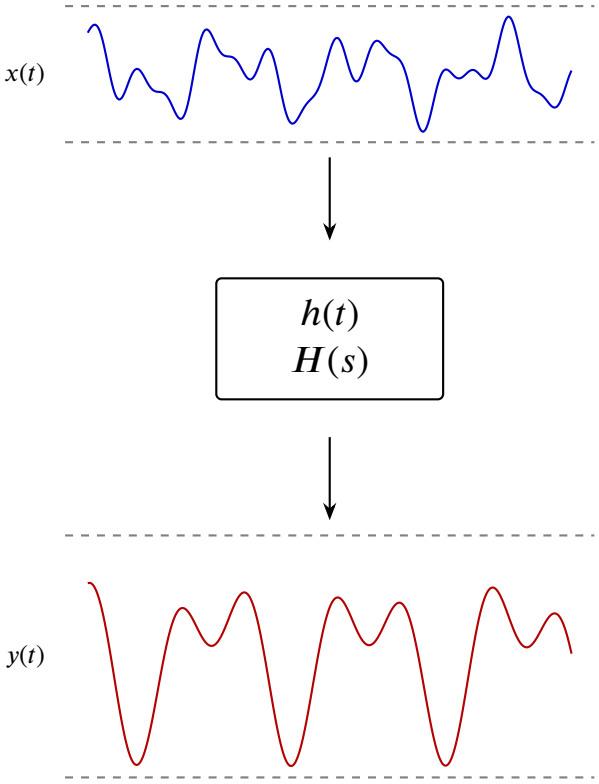
This only depends on $Re\{s\} = \sigma$, and ignore the $j\omega$ axis. When looking at RoC on a plane it is the strig parallel to $Im\{s\}$.

6.3 System Stability:

In a stable system, an input will not lead to an unbounded output. As such when $t \to \infty$, $y(t) \to 0$ or a value. An example is an RC circuit, when the capacitor is connected for long time, the capacitor reaches steady state V_s.

6.4 Bounded Input Bounded Output Stability:

A system is BIBO stable if every bounded input produces a bounded output.


x(t)
h(t)
H(s)
y(t)

6.5 Discrete Signal Stability:

- Consider a time signal where $|x[n]| < B_i$ for all n
- Output would become:

$$\begin{aligned} |y[n]| &= |\sum_{k=-\infty}^{\infty} h[k]x[n-k]| \\ &\leq \sum_{k=-\infty}^{\infty} |h[k]x[n-k]| \end{aligned}$$

Assuming maximum value for input for all n

$$|y[n]| \leq B_i \sum_{k=-\infty}^{\infty} |h[k]|$$

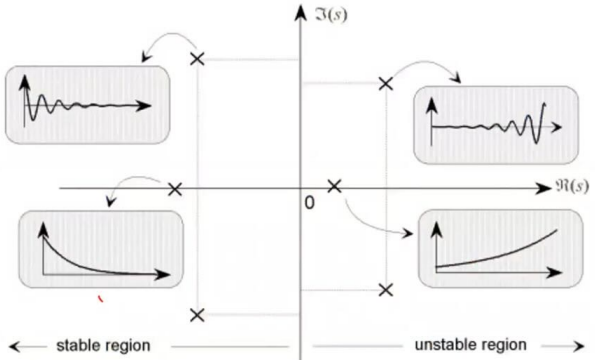
If input is bounded:

$$\sum_{k=-\infty}^{\infty} |h[k]| < \infty$$

6.6 System Stability - s-Domain:

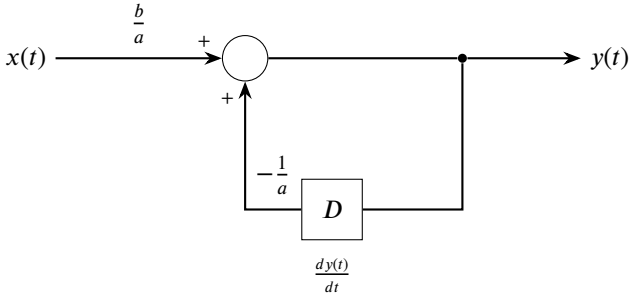
- Recall that the transfer function, $H(s)$, generates the transient response and input generates the steady state.
- For a system to be considered stable, all components must also be stable (bounded)
- Transfer function of LTI system are always rational
- Complex poles are always in conjugate pairs
- For stability, all poles of the transfer function must be on left plane

6.7 Location of Poles:



6.8 Block Diagrams:

Linear constant coefficient can be represented by block diagrams. This is useful for simulation or implementation of the system.



$$y(t) = \frac{b}{a}x(t) + (-\frac{1}{a})\frac{dy(t)}{dt}$$

$$\frac{dy(t)}{dt} + ay(t) = bx(t)$$

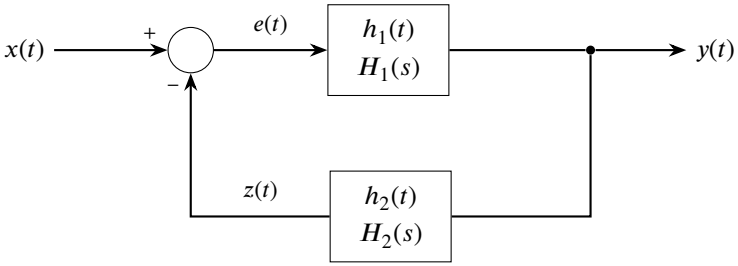
6.9 Using Integrators:

Differenntiators are often hard to implement and sensitive to errors. On the other hand integrators are easy to implement using OpAmps.

$$\frac{dy(t)}{dt} + ay(t) = bx(t)$$

$$y(t) = \int_{-\infty}^{t} bx(\tau) - ay(\tau) d\tau$$

6.10 Feedback System:



$$\begin{aligned}
Y(s) &= E(s)H(s) \\
E(s) &= X(s) - Z(s) \\
Z(s) &= H_2(s)Y(s) \\
Y(s) &= [X(s) - Z(s)]H_1(s) \\
&= [X(s) - H_2(s)Y(s)]H_1(s) \\
&= \frac{H_1(s)}{1 + H_1(s)H_2(s)} X(s)
\end{aligned}$$

6.11 z-Transform:

Similar to Laplace but for discrete time systems.

$$X[z] = \sum_{n=0}^{\infty} x[n] z^{-n}$$

z is a complex frequency variable $re^{j\omega}$. Example, Find the z-transform of $x[n] = a^n u[n]$

$$\begin{aligned} X[z] &= \sum_{n=0}^{\infty} a^n z^{-n} \\ &= \sum_{n=0}^{\infty} (az^{-1})^n \\ &= \lim_{n\to\infty} \frac{1 - az^{-1(n)}}{1 - az^{-1}} \\ &= \frac{z}{z-a} \end{aligned}$$

7 Fourier Series

7.1 Signals as Other Signals:

Any discrete time signal can be thought of as a sum of shifted and scaled impulse signals. The same can be true for continuous time signals. A periodic signal can be represented as a sum of sin and cos functions.

7.2 Periodic Waveforms:

These waveforms are defined by:

$$x(t) = x(t+T), \text{ where } t \in \mathbb{R}$$

Most natural waveforms are like sinusodial but not halfwave, AC Power, or square wave.

7.3 Fourier Series:

Any periodic series can be represented by an infinite sum of harmonically related series. The harmonis are integer multiplied by the fundamental frequency.

$$f(t) = a_v + \sum_{k=1}^{\infty} a_k \cos(k\omega_0 t) + b_k \sin(k\omega_0 t)$$

Where:

- $\omega = \frac{2\pi}{T}$
- $a_v = \frac{1}{T}\int_{t_0}^{t_0+T} f(t)dt$
- $a_k = \frac{2}{T}\int_{t_0}^{t_0+T} f(t)\cos(k\omega_0 t)dt$
- $b_k = \frac{2}{T}\int_{t_0}^{t_0+T} f(t)\sin(k\omega_0 t)dt$

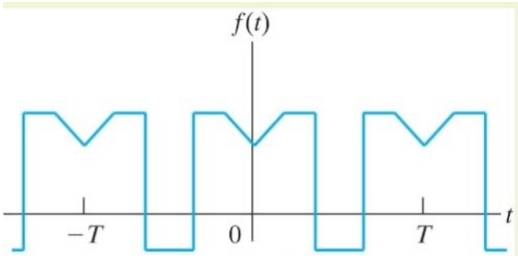
7.4 Dirichlet's Conditions:

4 items must be satisfied to a be Fouries transform:

1. Singled valued - when t only one value for $f(t)$
2. Have a finite number of discontinunity in period
3. Have a finite number of maxima and minima in one period
4. The $\int_{t_0}^{t_0+T} |f(t)|dt$ exist

7.5 Signal Symmetry and Fourier Coefficients:

7.5.1 Even



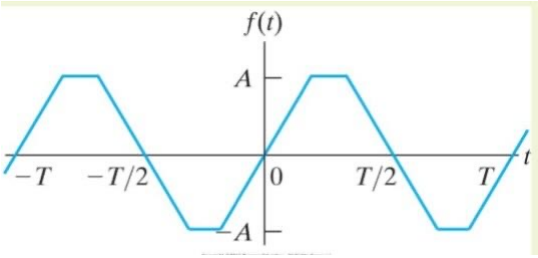
$$f(t) = f(-t)$$

$$a_v = \frac{2}{T}\int_0^{\frac{T}{2}} f(t)dt$$

$$a_k = \frac{4}{T}\int_0^{\frac{T}{2}} f(t)\cos(k\omega_0 t)dt$$

$$b_k = 0$$

7.5.2 Odd



$$f(t) = -f(-t)$$

$$a_v = 0$$
$$a_k = 0$$
$$b_k = \frac{4}{T}\int_0^{\frac{T}{2}} f(t)\sin(k\omega_0 t)dt$$

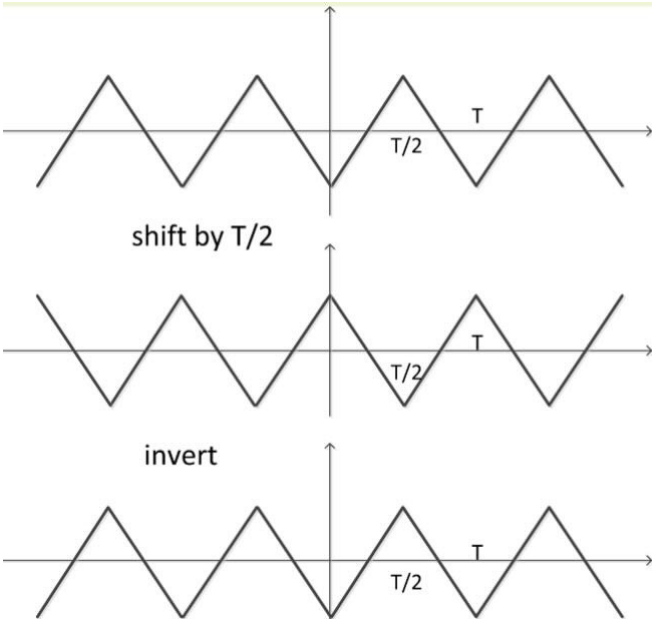
7.5.3 Non-symmetric

Shift signal until even or odd is found (T). Then shift resulting output.

$$\sin(k\omega_0[t+T])$$

$$\cos(k\omega_0[t+T])$$

7.6 Half-wave Symmetry:



$$f(t) = -f(t - \frac{T}{2})$$

$$a_v = 0$$

$$a_k = \begin{cases} 0, & k \text{ is even} \\ \frac{4}{T}\int_0^{\frac{T}{2}} f(t)\cos(k\omega_0 t)dt, & k \text{ is odd} \end{cases}$$

$$b_k = \begin{cases} 0, & k \text{ is even} \\ \frac{4}{T}\int_0^{\frac{T}{2}} f(t)\sin(k\omega_0 t)dt, & k \text{ is odd} \end{cases}$$

7.6.1 Even and Half-wave

$$a_v = 0$$

$$a_k = \begin{cases} 0, & k \text{ is even} \\ \frac{4}{T}\int_0^{\frac{T}{2}} f(t)\cos(k\omega_0 t)dt, & k \text{ is odd} \end{cases}$$

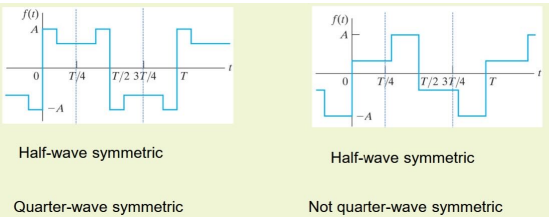
$$b_k = 0$$

7.6.2 Odd and Half-wave

$$a_v = 0$$
$$a_k = 0$$
$$b_k = \begin{cases} 0, & k \text{ is even} \\ \frac{4}{T}\int_0^{\frac{T}{2}} f(t)\sin(k\omega_0 t)dt, & k \text{ is odd} \end{cases}$$

7.7 Quater-wave Symmetry:

A signal that is half-wave symmetric and symmetric around the mid-point.



7.7.1 Even and Quater-wave

$$a_v = 0$$
$$a_k = \begin{cases} 0, & k \text{ is even} \\ \frac{8}{T}\int_0^{\frac{T}{4}} f(t)\cos(k\omega_0 t)dt, & k \text{ is odd} \end{cases}$$
$$b_k = 0$$

7.7.2 Odd and Quater-wave

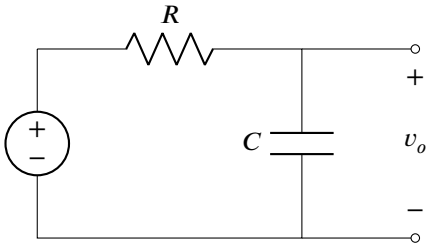
$$a_v = 0$$
$$a_k = 0$$
$$b_k = \begin{cases} 0, & k \text{ is even} \\ \frac{8}{T}\int_0^{\frac{T}{4}} f(t)\sin(k\omega_0 t)dt, & k \text{ is odd} \end{cases}$$

7.8 Steady-state Response:

A $\cos(\omega t + \theta)$ is applied with a transfer $H(s)$.

$$y(t) = A|H(j\omega)|\cos(\omega t + \varphi + \theta(\omega))v(t) + r(t)$$

$|H(j\omega)|$ is the steady-state response.



$$\begin{aligned}
v_g(t) &= \frac{4V_m}{\pi}\sum_{k=\text{odd}}^{\infty}\frac{\sin(k\omega_0 t)}{k} \\
H(s) &= \frac{1}{1+sRC} \\
H(j\omega) &= \frac{1}{1+j\omega RC} \\
|H(j\omega)| &= \frac{1}{\sqrt{1+\omega^2R^2C^2}} \\
&= \frac{1-j\omega RC}{1+\omega^2R^2C^2} \\
\angle H(j\omega) &= -\arctan(\omega RC)
\end{aligned}$$

$$v_0(t) = \frac{4V_m}{\pi}\sum_{k=\text{odd}}^{\infty}\frac{|H(j\omega k)|\sin(k\omega_0 t - \arctan[k\omega_0 RC])}{k}$$

As C increases the phase approaches 90° and magnitude is approximately $\frac{4V_m}{k^2\pi\omega_0 RC}$.

$$\begin{aligned}
v_0(t) &\approx \frac{4V_m}{\pi\omega_0 RC}\sum_{k=\text{odd}}^{\infty}\frac{\sin(k\omega_0 t - 90°)}{k^2} \\
&= \frac{-4V_m}{\pi\omega_0 RC}\sum_{k=\text{odd}}^{\infty}\frac{\cos(k\omega_0 t)}{k^2}
\end{aligned}$$

7.9 Fourier Series - Alternative Form:

$$z\cos(x-y) = z\cos(x)\cos(y) + z\sin(x)\sin(y)$$

$$\text{Let: } z\cos(y) = a \quad z\sin(y) = b$$

$$\begin{aligned} a^2 + b^2 &= z^2\cos^2(y) + z^2\sin^2(y) \\ &= z^2 \\ z &= \sqrt{a^2+b^2} \\ \frac{b}{a} &= \frac{z\sin(y)}{z\cos(y)} \\ &= \tan(y) \\ y &= \arctan(\frac{b}{a}) \end{aligned}$$

$$\begin{aligned} a\cos(x) + b\sin(x) &= z\cos(x-y) \\ &= \sqrt{a^2+b^2}\cos(x - \arctan(\frac{b}{a}) \end{aligned}$$

Fourier be written

$$\begin{aligned} f(t) &= a_v + \sum_{k=1}^{\infty} a_k\cos(k\omega_0 t) + b_k\sin(k\omega_0 t) \\ &= a_v\sum_{k=1}^{\infty}\sqrt{a_k^2+b_k^2}\cos(k\omega_0 t - \arctan(\frac{b_k}{a_k})) \end{aligned}$$

7.10 Average Power of Periodic Signal:

$$v(t) = v_{DC} + \sum_{k=1}^{\infty} v_k\cos(k\omega_0 t - \theta_{vk})$$

$$i(t) = i_{DC} + \sum_{k=1}^{\infty} i_k\cos(k\omega_0 t - \theta_{ik})$$

$$\begin{aligned} P_{av} &= \frac{1}{T}\int_{t_0}^{t_0+T} v(t)i(t)dt \\ &= v_{DC}i_{DC} + \sum_{k=1}^{\infty}\frac{v_k i_k}{2}\cos(\theta_{ik} - \theta_{vk}) \end{aligned}$$

7.11 RMS of Periodic Signals:

$$F_{RMS} = \sqrt{\frac{1}{T}\int_{t_0}^{t_0+T} f(t)^2 dt}$$

$$= \sqrt{\frac{1}{T}\int_{t_0}^{t_0+T} [f_{DC} + \sum_{k=1}^{\infty} f_k \cos^2(k\omega_0 t - \theta_{vk})dt}$$

$$= \sqrt{f_{DC}^2 + \sum_{k=1}^{\infty} f_{k(RMS)}^2}$$

7.12 Exponential Form of Fourier Series:

Using sin and cos identities in Euler form, it can be written in complex exponential.

$$\cos(\theta) = \frac{e^{j\theta} + e^{-j\theta}}{2}$$

$$\sin(\theta) = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$

$$f(t) = a_v + \sum_{k=1}^{\infty} a_k \cos(k\omega_0 t + b_k) + b_k \sin(k\omega_0 t)$$

$$= a_v + \sum_{k=1}^{\infty} a_k \frac{e^{jk\omega_0 t} + e^{-jk\omega_0 t}}{2} + b_k \frac{e^{jk\omega_0 t} - e^{-jk\omega_0 t}}{2j}$$

$$= a_v + \sum_{k=1}^{\infty} \frac{a_k - jb_k}{2} e^{jk\omega_0 t} + \frac{a_k + jb_k}{2} e^{-jk\omega_0 t}$$

Let: $c_k = \frac{a_k - jb_k}{2}$ & $\overline{c_k} = \frac{a_k + jb_k}{2}$

$$\therefore f(t) = a_v + \sum_{k=1}^{\infty} c_k e^{jk\omega_0 t} + \overline{c_k} e^{-jk\omega_0}$$

$$c_k = \frac{1}{T}\int_{t_0}^{t_0+T} f(t) e^{-jk\omega_0 t} dt$$

Take $k = 0$

$$c_0 = \frac{1}{T}\int_{t_0}^{t_0+T} f(t) dt$$

For $k < 0$

$$\begin{aligned} c_k &= \frac{1}{T}\int_{t_0}^{t_0+T} f(t)e^{jk\omega_0 t}dt \\ &= \overline{c_k} \end{aligned}$$

Combining all Coefficeints:

$$\begin{aligned} f(t) &= a_v + \sum_{k=1}^{\infty} c_k e^{jk\omega_0 t} + \overline{c_k}e^{-jk\omega_0 t} \\ &= c_0 + \sum_{k=1}^{\infty} c_k e^{jk\omega_0 t} + \sum_{k=1}^{\infty} \overline{c_k}e^{-jk\omega_0 t} \\ &= c_0 + \sum_{k=1}^{\infty} c_k e^{jk\omega_0 t} + \sum_{k=-1}^{-\infty} c_k e^{jk\omega_0 t} \\ &= \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t} \end{aligned}$$

7.13 Summary of Exponential Form:

$$f(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t}$$

$$c_k = \frac{1}{T}\int_{t_0}^{t_0+T} f(t)e^{-jk\omega_0 t}dt$$

$$F_{RMS} = \sqrt{c_0^2 + 2\sum_{k=1}^{\infty} |c_k|^2}$$

8 Fourier Transform

8.1 Fourier Series:

$$f(t) = \sum_{k=-\infty}^{\infty} c_k e^{j\omega_0 t}$$

$$c_k = \frac{1}{T}\int_{t_0}^{t_0+T} f(t)e^{-jk\omega_0 t}$$

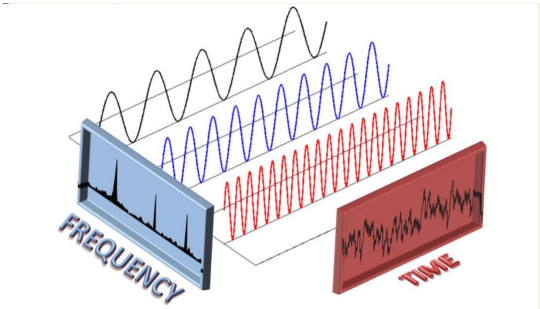
8.2 Fourier Transform:

As the period increases, the fundamental frequency decreases. When the period is infinite, spacing converges towards 0. This causes the summation to become an integral.

$$F(j\omega) = \int_{-\infty}^{\infty} f(t)e^{-j\omega t}dt$$

$$f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(j\omega)e^{j\omega t}d\omega$$

The Fourier tansofmr provides a frequency view of the function. Specifically showing frequencies of energy/power of signals contained.



$$\begin{aligned}
f(t) &= e^{-at}u(t) \\
F(j\omega) &= \int_{-\infty}^{\infty} e^{-at}u(t)e^{-j\omega t}dt \\
&= \int_{0}^{\infty} e^{-t(j\omega+a)}dt \\
&= \left[\frac{-1}{j\omega+a}e^{-t(j\omega+a)}\right]_0^{\infty} \\
&= \left[\frac{-1}{j\omega+a}e^{-\infty}\right] - \left[\frac{-1}{j\omega+a}\right] \\
&= \frac{1}{j\omega+a} \\
|F(j\omega)| &= \frac{1}{\sqrt{\omega^2+a^2}} \\
\angle F(j\omega) &= -\arctan\left(\frac{\omega}{a}\right)
\end{aligned}$$

8.3 Inverse Fourier Transform:

Unlike Laplace transforms, the inverse fourier is quite simple.

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(j\omega) e^{j\omega t} d\omega$$

Similar to Laplace it is easier to do it in partial fractions.

8.4 Laplace vs. Fourier:

The fourier transform is a special case of the laplace transform. If all poles are on the left-side of the s-plane, the Fourier series converges. Then substitute $s = j\omega$.

- If $f(t) = 0$ for $t \leq 0$, $\mathcal{F}\{f(t)\} = \mathcal{L}\{f(t)\}|_{s=j\omega}$.
- Else $f(t) = 0$ for $t \geq 0$, $\mathcal{F}\{f(t)\} = \mathcal{L}\{f(t)\}|_{s=j\omega}$.
- Else $f(t) = a \; \forall \; t \in (-\infty, \infty)$:

$$f(t) = f(t)^{+} + f(t)^{-}$$

$$F(j\omega) = \mathcal{L}\{f(t)^{+}\}|_{s=j\omega} + \mathcal{L}\{f(-t)^{-}\}|_{s=-j\omega}$$

8.5 Convergence of Fourier Integral:

Must meet the following conditions:

1. Single valued
2. Have a finite number of discontinuities within any finite interval
3. Have finite maxima and minima within finite interval
4. $\int_{-\infty}^{\infty} |f(t)| dt$ exists

An example of constant value A

$$
\begin{aligned}
f(t) &= A \\
&= Ae^{-\epsilon|t|} \\
\text{When } \epsilon \to 0 \quad & f(t) \to A \\
F(j\omega) &= \frac{2\epsilon A}{\epsilon^2 + \omega^2} \\
F(0) &= \frac{2A}{\epsilon} \\
\text{At } \omega \neq 0,\ & F(j\omega) \to 0 \text{ when } \epsilon \to 0 \\
\int_{-\infty}^{\infty} \frac{2\epsilon A}{\epsilon^2 + \omega^2} d\omega &= \left[2\epsilon A \frac{1}{\epsilon} \arctan\left(\frac{\omega}{a}\right)\right]_{-\infty}^{\infty} \\
&= 2\pi A
\end{aligned}
$$

$$
\begin{aligned}
\therefore f(t) &= A \\
F(j\omega) &= 2\pi A\delta(\omega)
\end{aligned}
$$

8.6 Fourier Transform of Periodic:

$$
\begin{aligned}
\mathcal{F}\{\cos(\omega_0 t)\} &= \pi\delta(\omega - \omega_0) + \pi\delta(\omega - \omega_0) \\
\mathcal{F}\{\sin(\omega_0 t)\} &= j\pi\delta(\omega - \omega_0) - j\pi\delta(\omega + \omega_0) \\
\mathcal{F}\{e^{j\omega_0 t}\} &= 2\pi\delta(\omega - \omega_0)
\end{aligned}
$$

The fourier tansform of any periodic signal can be experssed as an infinite of impulse functions.

$$f(t) = \sum_{k=-\infty}^{\infty} c_k e^{j\omega_0 kt}$$

$$F(j\omega) = \sum_{k=-\infty}^{\infty} 2\pi c_k \delta(\omega - k\omega_0)$$

8.7 Operational Transforms:

8.7.1 Linearity

$$\mathcal{F}\{k_1 f_1(t) \pm k_2 f_2(t)\} = k_1 F_1(j\omega) \pm k_2 F_2(j\omega)$$

8.7.2 Differentiation

$$\mathcal{F}\left\{\frac{d^n f(t)}{dt^n}\right\} = (j\omega)^n F(j\omega)$$

8.7.3 Integration

$$\mathcal{F}\left\{\int_{-\infty}^{t} f(\tau)d\tau\right\} = frac F(j\omega)j\omega + \pi F(0)\delta(\omega)$$

8.7.4 Translation in t-Domain

$$\mathcal{F}\{f(t-t_0)\} = e^{-j\omega t_0}F(j\omega)$$

8.7.5 Translation in w-Domain

$$\mathcal{F}\{e^{j\omega_0 t}f(t)\} = F(j[\omega-\omega_0])$$

8.7.6 Scale Change

$$\mathcal{F}\{f(at)\} = \frac{1}{|a|}F(\frac{j\omega}{a})$$

8.7.7 Convolution in t-Domain

$$\mathcal{F}\{f(t)\otimes g(t)\} = F(j\omega)G(j\omega)$$

8.7.8 Convolution in w-Domain

$$\mathcal{F}\{f(t)g(t)\} = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(j\theta)G(j[\omega-\theta])$$

8.7.9 Modulation

$$\mathcal{F}\{f(t)\cos(\omega_0 t)\} = \frac{1}{2}[F(\omega-\omega_0)+F(\omega+\omega_0)]$$

8.8 Conjugate Symmetry:

$$\begin{aligned}
\mathcal{F}\{f(t)\} &= F(j\omega) \\
\mathcal{F}\{\overline{f(t)}\} &= F(-j\omega) \\
\text{If } f(t) &= \overline{f(t)} \\
F(j\omega) &= \overline{F(-j\omega)} \\
\overline{F(j\omega)} &= F(-j\omega)
\end{aligned}$$

$$\begin{aligned}
F(j\omega) &= \Re\{F(j\omega)\} + j\Im\{F(j\omega)\} \\
\overline{F(j\omega)} &= \Re\{F(j\omega)\} - j\Im\{F(j\omega)\} \\
&= \Re\{F(-j\omega)\} + j\Im\{F(-j\omega)\}
\end{aligned}$$

If $f(t)$ is Real

$$\begin{aligned}
\Re\{F(j\omega)\} &= \Re\{F(-j\omega)\} \\
\Im\{F(j\omega)\} &= -\Im\{F(-j\omega)\}
\end{aligned}$$

Express $F(j\omega$ in polar form

$$\begin{aligned}
F(j\omega) &= |F(j\omega)|e^{j\angle F(j\omega)} \\
|F(j\omega)| &= \sqrt{\Re\{F(j\omega)\}^2 + \Im\{F(j\omega)\}^2} \\
\angle F(j\omega) &= \arctan\left(\frac{\Im\{F(j\omega)\}}{\Re\{F(j\omega)\}}\right)
\end{aligned}$$

If $f(t)$ is real and even:

$$\begin{aligned}
F(j\omega) &= \int_{-\infty}^{\infty} f(t)e^{-j\omega t}dt \\
\overline{F(j\omega)} &= F(-j\omega) \\
&= \int_{-\infty}^{\infty} f(t)e^{j\omega t}dt \\
\text{Sub } t &= -\tau \\
F(-j\omega) &= -\int_{\infty}^{-\infty} f(-\tau)e^{-j\omega\tau}d\tau \\
&= F(j\omega) \\
&= \overline{F(j\omega)}
\end{aligned}$$

If $f(t)$ is real and even, $F(j\omega)$ is real and even. Else $f(t)$ is real and odd, $F(j\omega)$ is imaginary and odd.

8.9 Duality Principle:

Between a Fourier Transform and its inverse are similar when applied in pairs. Generally:

$$\mathcal{F}\{f(t)\} = F(\omega)$$

$$\mathcal{F}\{F(\omega)\} = 2\pi f(-\omega)$$

8.10 Parseval's Theorem:

If $f(t)$ and $F(j\omega)$ are Fourier pairs,

$$\int_{-\infty}^{\infty} |f(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |F(j\omega)|^2 d\omega$$

The total energy within a signal can be found with either side of the above equation. As such $|F(j\omega)|^2$ is referred to the energy density spectrum. Note that if a finite amount of energy is calculated over an infinite interval, the power is 0.

9 Frequency Selective Circuits

9.1 Time vs. Frequency Domain:

Usually signals are most intuative in t-Domain (e.g. Audio, Power, or Control Systems). However, sometimes frequency-domain is intuative (e.g. Mixed Noise Signals). In this case the fouries transform shows how the signal energy is spread throughout the freqencies.

9.2 Signal Transmission:

To transmit a signal, all frequencies must be transmitted. This requiries a large amount of bandwidth, increasing the overall cost of the transmission. By finding where the energy is concentrated, only this part of the signal can be sent, reducing the bandwidth requirement.

9.3 Circuit Components and Frequency:

The impedance of certain circuit components varies with frequencies. These include capacitors, inductors and diodes.

9.4 Effects of Phase Changes:

- Effects of magnitude change with many frequencies are easier to identify.
- Phase change can have a significant effect on output but harder to see.

9.5 Frequency Response of an LTI System:

$$\begin{aligned} H(j\omega) &= \frac{V_{out}(j\omega)}{V_{in}(j\omega)} \\ &= |H(j\omega)|e^{j\angle H(j\omega)} \\ &= |H(j\omega)|\angle H(j\omega) \end{aligned}$$

9.6 Effects of Phase Response:

Linear Phase - Has the following response:

$$\begin{aligned} H(j\omega) &= e^{-j\omega t_0} \\ |H(j\omega)| &= 1 \\ \angle H(j\omega) &= -\omega t_0 \end{aligned}$$

This system has a linear phase response, the output:

$$\begin{aligned} V_o(j\omega) &= H(j\omega)V_i(j\omega) \\ &= e^{-j\omega t_0}V_i(j\omega) \\ v_o(t) &= v_i(t-t_0) \\ \mathcal{F}\{f(t-t_0)\} &= e^{-j\omega t_0}F(j\omega) \end{aligned}$$

9.7 Bode Plot:

The frequency response of a system is understood better with a plot of it Magnitude and angle (in degrees) responses. This is usually done with a logarithmic scale.

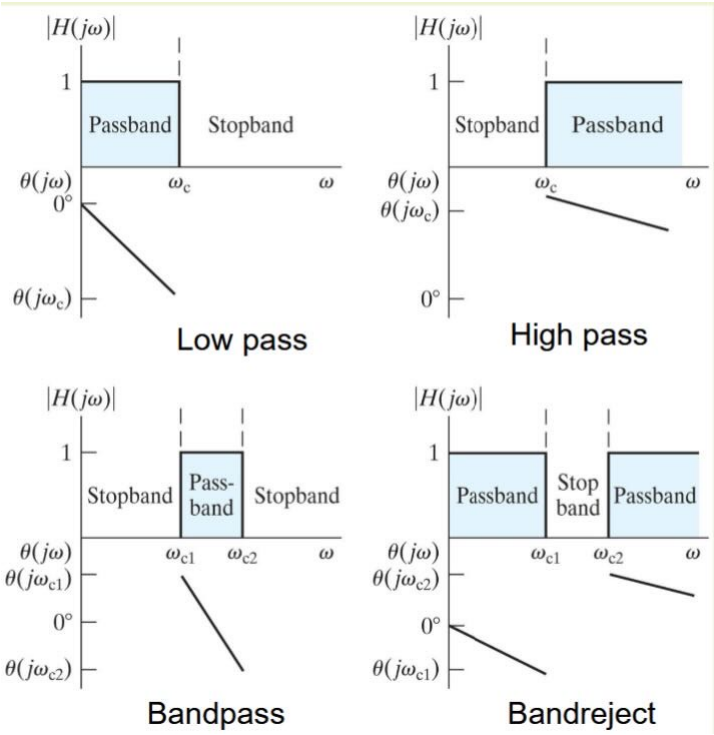
$$dB = 20\log_{10}(|H(j\omega)|)$$

9.8 Frequency Selective Circuits:

System impedance for some circuits allow a high frequency to product a low amplitude. This allows certain frequencies to be passed or blocked. Some of these filter circuits include:

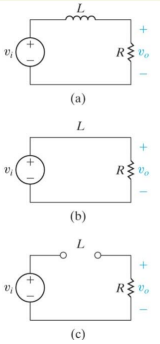
- Low Pass
- High Pass
- Band-pass
- Band-reject (Notch)

9.9 Filter Categories - Ideal:



9.9.1 Low Pass

- Series RL circuit can behave as low pass filter
- Inductor impedance Z=jωL. At low frequencies inductor impedance is lower compared to R (short circuit)
- Then v_o=v_i
- $H(j\omega) = 1$
- At high frequencies inductor impedance is larger compared to R (open circuit)
- Then v_o=0
- $H(j\omega) = 0$



Using voltage divider:

$$
\begin{aligned}
H(j\omega) &= \frac{V_o(j\omega)}{V_i(j\omega)} \\
&= \frac{R}{R + j\omega L} \\
&= \frac{\frac{R}{L}}{j\omega + \frac{R}{L}} \\
|H(j\omega)| &= \frac{\frac{R}{L}}{\sqrt{\omega^2 + \left(\frac{R}{L}\right)^2}} \\
\angle H(j\omega) &= -\arctan\left(\frac{\omega L}{R}\right)
\end{aligned}
$$

Cut off Frequency defines the frequency where the average power is $\frac{1}{2}$ of Max:

$$P_{av} = \frac{1}{2}\frac{V_o^2}{R}$$
$$P_{av(max)} = \frac{1}{2}\frac{V_{o(max)}^2}{R}$$
$$P_{av(\omega_c)} = \frac{1}{2}\frac{V_{o(\omega_c)}^2}{R}$$
$$V_{o(\omega_c)}^2 = \frac{1}{2}V_{o(max)}^2$$
$$|H(j\omega_c)|^2 V_i^2 = \frac{1}{2}|H(j\omega)_{max}|^2 V_i^2$$
$$|H(j\omega_c)| = \frac{1}{\sqrt{2}}|H(j\omega)_{max}|$$

Calculating $H(j\omega)_{max}$ is at $\omega = 0$

$$|H(j\omega)_{max}| = \frac{\frac{R}{L}}{\sqrt{\frac{R}{L}^2}} = 1$$

Cut off it at $H(j\omega_c) = \frac{1}{\sqrt{2}}$

$$\frac{\frac{R}{L}}{\sqrt{\omega_c^2 + \left(\frac{R}{L}\right)^2}} = \frac{1}{\sqrt{2}}$$
$$\omega_c^2 + \left(\frac{R}{L}\right)^2 = 2\left(\frac{R}{L}\right)^2$$
$$\omega_c = \frac{R}{L}$$

The transfer function is best done in s-Domain, even if designed in frequency or t-Domain.

Sub $j\omega = s$

$$H(j\omega) = \frac{\frac{R}{L}}{j\omega + \frac{R}{L}}$$

$$H(s) = \frac{\frac{R}{L}}{s + \frac{R}{L}}$$

Since $\frac{R}{L} = \omega_c$

$$H(s) = \frac{\omega_c}{s + \omega_c}$$

Cut off magnitude is dB is 0 since the maximum magnitude is 1. At cutoff magnitude is $\frac{1}{\sqrt{2}}$, the dB is -3. In Bode plots, the cut off is identified as magnitude is 3dB below maximum magnitude.
Rolloff is the slope of the magnitude response as it drops after the cut off frequency. Usually this is expressed in dB/decade.

$$|H(j\omega)| = \frac{\omega_c}{\sqrt{\omega^2 + \omega_c^2}}$$

$$|H(j10\omega)| = \frac{\omega_c}{\sqrt{10^2\omega^2 + \omega_c^2}}$$

For frequencies where $\omega \gg \omega_c$:

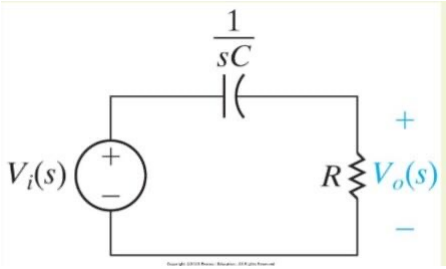
$$|H(j\omega)| = \frac{\omega_c}{\omega}$$

$$|H(j10\omega)| = \frac{\omega_c}{10\omega}$$

Converting the different to dB

$$20\log_{10}\left(\frac{\omega_c}{\omega}\right) - 20\log_{10}\left(\frac{\omega_c}{10\omega}\right)$$
$$=20\log_{10}(10)$$
$$=20\text{dB/decade}$$

9.9.2 High Pass Filter



$$
\begin{aligned}
H(j\omega) &= \frac{V_o(j\omega)}{V_i(j\omega)} \\
&= \frac{R}{R + \frac{1}{j\omega C}} \\
&= \frac{j\omega}{j\omega + \frac{1}{RC}}
\end{aligned}
$$

Magnitude Response:

$$
|H(j\omega)| = \frac{\omega}{\sqrt{\omega^2 + \left(\frac{1}{RC}\right)^2}}
$$

Phase Response:

$$
\angle H(j\omega) = \arctan\left(\frac{1}{\omega RC}\right) = 90° - \arctan(\omega RC)
$$

Cut off Frequency and Transfer Function:

$$H(j\omega)_{max} = 1$$

$$\text{Then } H(j\omega_c) = \frac{1}{\sqrt{2}}$$

$$\frac{\omega}{\sqrt{\omega^2 + \left(\frac{1}{RC}\right)^2}} = \frac{1}{\sqrt{2}}$$

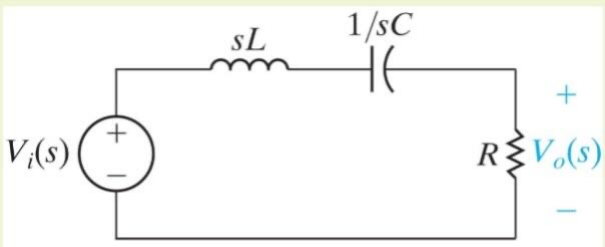
$$\omega_c^2 + \left(\frac{1}{RC}\right)^2 = 2\omega_c^2$$

$$\omega_c = \frac{1}{RC}$$

$$H(j\omega) = \frac{j\omega}{j\omega + \frac{1}{RC}}$$

$$H(s) = \frac{s}{s + \omega_c}$$

9.9.3 Band-pass Filter



$$H(j\omega) = \frac{V_o(j\omega)}{V_i(j\omega)}$$

$$= \frac{R}{R + j\omega L + \frac{1}{j\omega C}}$$

$$= \frac{j\omega\frac{R}{L}}{-\omega^2 + \frac{1}{LC} + j\omega\frac{R}{L}}$$

Magnitude Response:

$$h(j\omega)| = \frac{\omega\frac{R}{L}}{\sqrt{\left(\frac{1}{LC} - \omega^2\right)^2 + \left(\frac{\omega R}{L}\right)^2}}$$

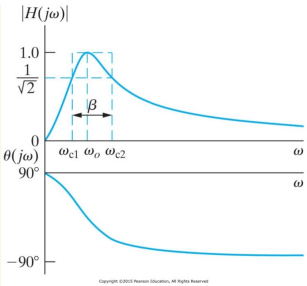
Phase Response:

$$\angle H(j\omega) = \arctan\left(\frac{\frac{1}{LC} - \omega^2}{\frac{\omega R}{L}}\right) = 90° - \arctan\left(\frac{\frac{\omega R}{L}}{\frac{1}{LC} - \omega^2}\right)$$

Two cutoff frequencies

Frequency range between the two cutoff frequencies is the bandwidth, β, of the filter.

Frequency where the gain is maximum is the centre frequency, ω_0. This is the resonant frequency of the circuit where the impedance is purely real



9.9.4 Band-pass Characteristics

At the centre frequency, the capacitor and inductor impendance cancel each other out.

$$H(j\omega_0) = \frac{R}{R + j\omega_0 L + \frac{1}{j\omega_0 C}} = \frac{R}{R} = 1$$

When $|H(j\omega)_{max}| = 1$:

$$j\omega_0 L - \frac{j}{\omega_0 C} = 0$$

$$\omega_0 = \sqrt{\frac{1}{LC}}$$

Cut off Frequency:

$$\begin{aligned} |H(j\omega_c)| &= \frac{\frac{\omega_c R}{L}}{\sqrt{\left(\frac{1}{LC} - \omega_c^2\right) + \left(\frac{\omega_c R}{L}\right)^2}} \\ &= \frac{1}{\sqrt{2}} \end{aligned}$$

$$\begin{aligned} \therefore \omega_c^2 L \pm \omega_c R - \frac{1}{C} &= 0 \\ \omega_{c1,2} &= \pm\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} \\ \omega_{c1}\omega_{c2} &= \omega_0^2 \\ \therefore \omega_0 &= \sqrt{\omega_{c1}\omega_{c2}} \end{aligned}$$

Bandwidth, β:

$$\begin{aligned} \beta &= |\omega_{c1} - \omega_{c2}| \\ &= \frac{R}{L} \end{aligned}$$

Quality factor, Q, is the width of the bandpass independent of the center frequency.

$$Q = \frac{\omega_0}{\beta} = \frac{\sqrt{\frac{1}{LC}}}{\frac{R}{L}}$$

$$= \frac{1}{R}\sqrt{\frac{L}{C}}$$

$$H(j\omega) = \frac{j\omega\frac{R}{L}}{-\omega^2 + j\omega\frac{R}{L} + \frac{1}{LC}}$$

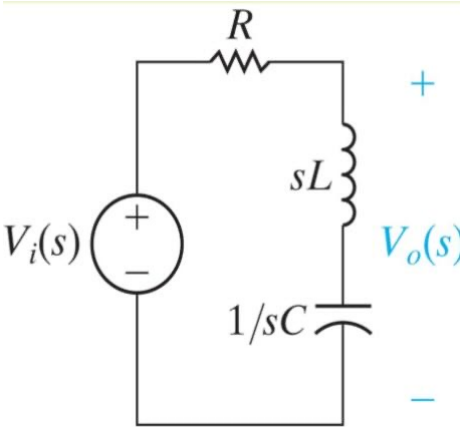
$$H(s) = \frac{s\frac{R}{L}}{s^2 + s\frac{R}{L} + \frac{1}{LC}}$$

$$\omega_0 = \sqrt{\frac{1}{LC}}$$

$$\beta = \frac{R}{L}$$

$$H(s) = \frac{s\beta}{s^2 + s\beta + \omega_0^2}$$

9.9.5 Band-reject Filter



$$H(j\omega) = \frac{j\omega L + \frac{1}{j\omega C}}{R + j\omega L + \frac{1}{j\omega C}}$$
$$= \frac{-\omega^2 + \frac{1}{LC}}{-\omega^2 + j\omega\frac{R}{L} + \frac{1}{LC}}$$

Magnitude Response:

$$|H(j\omega)| = \frac{|\frac{1}{LC} - \omega^2|}{\sqrt{\left(\frac{1}{LC} - \omega^2\right)^2 + \left(\frac{\omega R}{L}\right)^2}}$$

Phase Response:

$$\angle H(j\omega) = \arctan\left(\frac{\frac{-\omega R}{L}}{\frac{1}{LC} - \omega^2}\right)$$
$$= -\arctan\left(\frac{\frac{\omega R}{L}}{\frac{1}{LC} - \omega^2}\right)$$

Cut off Frequency:

$$\omega_{c1,2} = \pm\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \left(\frac{1}{LC}\right)}$$

Center Frequency:

$$\omega_c = \sqrt{\frac{1}{LC}}$$

Bandwidth:

$$\beta = \frac{R}{L}$$

Q Factor:

$$Q = \frac{1}{R}\sqrt{\frac{L}{C}}$$

Transfer Function:

$$H(j\omega) = \frac{-\omega^2 + \frac{1}{LC}}{-\omega^2 + j\omega\frac{R}{L} + \frac{1}{LC}}$$

$$H(s) = \frac{s + \frac{1}{LC}}{s^2 + s\frac{R}{L} + \frac{1}{LC}}$$

$$\omega_0 = \sqrt{\frac{1}{LC}}$$

$$\beta = \frac{R}{L}$$

$$H(s) = \frac{s + \omega_0^2}{s + s\beta + \omega_0^2}$$

10 Active Filter Design

10.1 Filter Design:

A filter can be designed when the caracteristic parameters of the filter are provided:

- Type
- Center Frequency
- Cut off frequencies
- Q Factor
- Bandwidth
- Magnitude and Phase Responses

As all parameters are inter-related, not all need to be specified.

10.2 Filter Gain:

- Some applications need the filters to select and amplify certain frequencies.

- This requires the magnitude response to be greater than 1.
- One method of adding gain is to have the filter combined with an OpAmp.

Low Pass	High Pass
$H(s) = \frac{K\omega_c}{s+\omega_c}$	$H(s) = \frac{Ks}{s+\omega_c}$
Bandpass	**Band reject**
$H(s) = \frac{Ks\beta}{s^2+s\beta+\omega_0^2}$	$H(s) = \frac{K(s^2\omega_0^2}{s^2+s\beta+\omega^2}$

10.3 Filter Order:

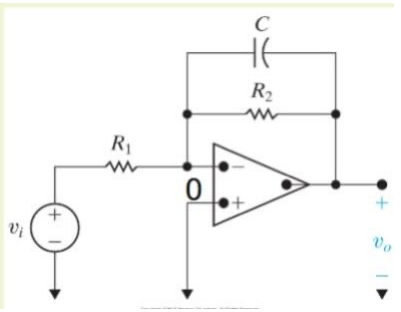
For continuous time filters, the number of poles of the filter $H(s)$ is the "order of the filter". As the filter order increases it becomes closer to the ideal characteristics of the filter.

10.4 Active Filters:

Analogue filters include active devices, such as amplifiers. Common active filters use OpAmps. They include advantages such as,

- Flexibility to adjust frequency and gain,
- Can aviod using inductors, and
- No loading issues.

10.4.1 First Order Active Low Pass



$$\frac{0-V_i}{R_1}+\frac{0-V_0}{R_2}+\frac{0-V_0}{\frac{1}{sC}}=0$$

$$V_0\left(\frac{1}{R_2}+sC\right)=\frac{-V_i}{R_1}$$

$$H(s)=\frac{V_0}{V_i}=\frac{-\frac{1}{R_1}}{\frac{1}{R_2}+sC}$$

$$=\frac{-\frac{1}{R_1}}{\frac{1+sR_2C}{R_2}}$$

$$=\frac{-R_2}{R_1(1+sR_2C)}$$

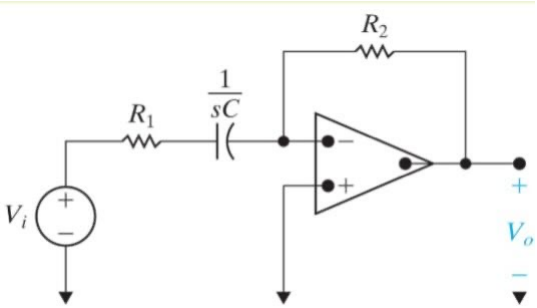
$$=\frac{\frac{R_2}{R_1}\frac{1}{R_2C}}{s+\frac{1}{CR_2}}$$

$$\text{Gain}=K=\frac{R_2}{R_1}$$

$$\omega_c=\frac{1}{CR_2}$$

The negative sign indicateds the phase will be inverted when compared to the passive design.

10.4.2 First Order Active High Pass



$$\frac{0-V_i}{R_1+\frac{1}{sC}}+\frac{0-V_o}{R_2}=0$$

$$V_o\left(\frac{1}{R_2}\right)=\frac{-V_i}{R_1+\frac{1}{sC}}$$

$$H(s)=\frac{V_o}{V_i}=\frac{s\frac{R_2}{R_1}}{s+\frac{1}{sR_1C}}$$

$$\text{Gain}=K=\frac{R_2}{R_1}$$

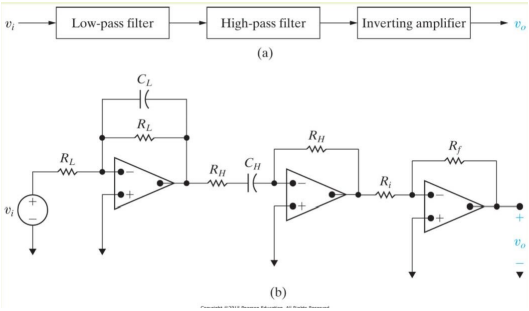
$$\omega_c=\frac{1}{CR_1}$$

10.4.3 Active Bandpass Filter

A bandpass filter frequency response can be though of as consisting of 3 seperate responses:

1. Unity gain low passs with ω_{c2}
2. Unity gain high pass with ω_{c1}
3. Gain provided by seperate amplifier

Hence a bandpass filter can be constructed by cascading a low pass, high pass and amplifier together in series. For the $\omega_{c2} \geq 2\omega_{c1}$. A filter with this frequency characteristic are called broadband filters.



$$
\begin{aligned}
H(s) &= \left(\frac{-\omega_{c2}}{s+\omega_{c2}}\right)\left(\frac{-s}{s+\omega_{c1}}\right)\left(\frac{-R_f}{R_i}\right) \\
&= \frac{K\omega_{c2}s}{(s+\omega_{c2})(s+\omega_{c1})} \\
&= \frac{K\omega_{c2}s}{s^2+(\omega_{c2}+\omega_{c1})s+\omega_{c1}\omega_{c2}}
\end{aligned}
$$

If $\omega_{c2} \gg \omega_{c1}$, then:

$$
\omega_{c2} - \omega_{c1} \approx \omega_{c2}
$$

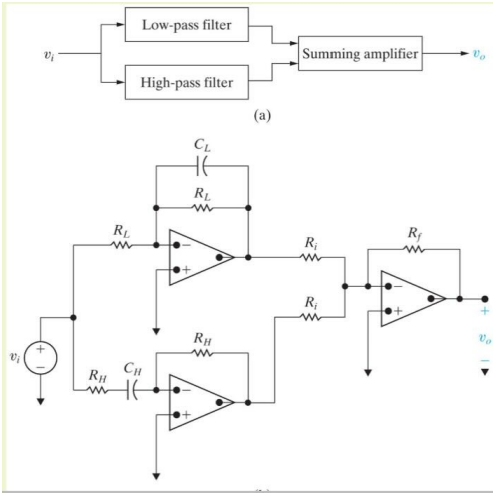
$$
H(s) = \frac{K\omega_{c2}s}{s^2+\omega_{c2}s+\omega_{c1}\omega_{c2}}
$$

10.4.4 Active Bandreject Filters

Similar to the bandpass filter, this can be thought of in 2 stages:

- Unity gain low pass with ω_{c1}
- Unity gain high pass with ω_{c2}, in parallel in with the low pass
- Gain by seperate amplifier

This filter cannot be cascading. As such the filters are in parrallel.



$$H(s) = \left(\frac{-\omega_{c1}}{s + \omega_{c1} + \frac{-s}{s+\omega_{c2}}} \right) \left(\frac{-R_f}{R_i} \right)$$
$$= -K \frac{-\omega_{c1}s - \omega_{c1}\omega_{c2} - s^2 - \omega_{c1}s}{(s+\omega_{c2})(s+\omega_{c1})}$$
$$= K \frac{s^2 + 2\omega_{c1}s + \omega_{c1}\omega_{c2}}{s^2 + (\omega_{c2} + \omega_{c1})s + \omega_{c1}\omega_{c2}}$$

Assuming $\omega_{c2} \gg \omega_{c1}$, then:

$$\omega_{c2} - \omega_{c1} \approx \omega_{c2}$$

$$H(s) = K \frac{s^2 + 2\omega_{c1}s + \omega_{c1}\omega_{c2}}{s^2 + \omega_{c2}s + \omega_{c1}\omega_{c2}}$$

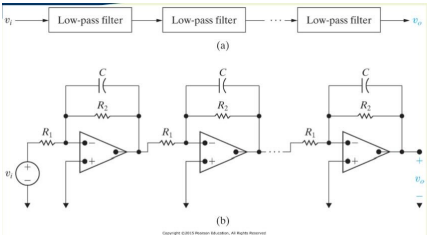
10.5 Cascading Filters:

Since, the circuis is in s-Domain, each transfer function is multiplied:

$$H(s) = H_1(s)H_2(s)H_3(s)\ldots$$

$$|H(j\omega)| = |H_1(j\omega)||H_2(j\omega)||H_3(j\omega)|\ldots$$
$$\angle H(j\omega) = \angle H_1(j\omega) + \angle H_2(j\omega) + \angle H_3(j\omega) + \ldots$$

10.6 n Order Low Pass Filter:



$$
\begin{aligned}
H(s) &= \left(\frac{-K\omega_c}{s+\omega_c}\right)^n \\
H(j\omega) &= \left(\frac{-K\omega_c}{j\omega+\omega_c}\right)^n \\
|H(j\omega)| &= \left|\frac{K\omega_c}{j\omega+\omega_c}\right|^n \\
&= \frac{(K\omega_c)^n}{\left(\sqrt{\omega_c^2+\omega^2}\right)^2} \\
\angle H(j\omega) &= -n \times \arctan\left(\frac{\omega}{\omega_c}\right)
\end{aligned}
$$

Cut off Frequency:
For low ω the gain is K^n. Then at cut off gain is $\frac{K^n}{\sqrt{2}}$.

$$
\begin{aligned}
|H(j\omega_{cn})| = \frac{K^n}{\sqrt{2}} &= \frac{(K\omega_c)^n}{\left(\sqrt{\omega_c^2+\omega_{cn}^2}\right)^n} \\
\sqrt{\omega_c^2+\omega_{cn}^2} &= \sqrt[n]{2}\omega_c \\
\omega_{cn} &= \left(\sqrt{\sqrt[n]{2}-1}\right)\omega_c
\end{aligned}
$$

Roll off:
the difference in dB gain aat the frequency $\omega \gg \omega_c$.

$$
\begin{aligned}
&20\log_{10}\left(\frac{\omega_{cn}}{\omega}\right) - 20\log_{10}\left(\frac{\omega_{cn}}{10\omega}\right)^n \\
=&20n\log_{10}(10) \\
=&20n \text{ dB/decade}
\end{aligned}
$$

10.7 Unity Design and Scaling:

For simplicity it is usually assumed that ω_c for filters is 1rad/s. And then these values are achieved throuhg using component vlaues of 1Ω, $1F$, and $1H$. The

actual values are these values scaled according to the actual values specified.

$$R' = k_m R$$
$$L' = \frac{k_m}{k_f} L$$
$$C' = \frac{1}{k_m k_f} C$$

10.8 Filter Gain around Cutoff:

Consider the gain of an n-stage low pass filter.

$$|H(j\omega)| = \frac{(\omega_{cn})^n}{\left(\sqrt{\omega_{cn}^2 + \omega^2}\right)^n}$$

At low frequency, $\omega \ll \omega_{cn}$, the gain is 1. When $\omega \to \omega_{cn}$.

$$|H(j\omega)| = \frac{1}{\left(\sqrt{1 + \frac{\omega^2}{\omega_{cn}^2}}\right)^n} < 1$$

Hence around cut off, the gain drops below 1. This make the filter less sharp around ω_c.

10.9 Butterworth Filter:

Butterworth filters are designed to provide a gain of 1, very close to the cut off frequency. As such the transfer function of the butterworth filter low of order n.

$$|H(j\omega)| = \frac{1}{\sqrt{1 + \left(\frac{\omega}{\omega_c}\right)^{2n}}}$$

- The cut off frequency is independent of the filter order
- For large n gin is close to unity for $\omega < \omega_c$
- Exponent of $\frac{\omega}{\omega_c}$ is always even.

$$
\begin{aligned}
|H(j\omega)|^2 &= H(j\omega)\overline{H(j\omega)} \\
&= H(j\omega)H(-j\omega) \\
&= H(s)H(-s) \\
&= \frac{1}{1+\omega^{2n}}
\end{aligned}
$$

$$
\begin{aligned}
H(s)H(-s) &= \frac{1}{1+(-s^2)^n} \\
&= \frac{1}{1+(-1)^n s^{2n}}
\end{aligned}
$$

The poles of a butterworth are $1+(-1)^n s^{2n}$.

TABLE 15.1 Normalized (so that $\omega_c = 1$ rad/s) Butterworth Polynomials up to the Eighth Order

n	nth-Order Butterworth Polynomial
1	$(s+1)$
2	$(s^2+\sqrt{2}s+1)$
3	$(s+1)(s^2+s+1)$
4	$(s^2+0.765s+1)(s^2+1.848s+1)$
5	$(s+1)(s^2+0.618s+1)(s^2+1.618s+1)$
6	$(s^2+0.518s+1)(s^2+\sqrt{2}+1)(s^2+1.932s+1)$
7	$(s+1)(s^2+0.445s+1)(s^2+1.247s+1)(s^2+1.802s+1)$
8	$(s^2+0.390s+1)(s^2+1.111s+1)(s^2+1.6663s+1)(s^2+1.962s+1)$